USED-CAR RESALE PRICE PROJECT

Data Dictionary & Dataset Overview | Phase 1: Data Understanding

1. Dataset overview

Dataset source	Kaggle: sujay1844/used-car-prices
File loaded	train.csv
Dataset dimensions	5,847 observations (rows) × 14 columns
Project objective	Understand the attributes that may influence the resale/listing price of a used car.
Target variable	Price

2. Data dictionary

Descriptions below are based on the supplied notebooks and the fields in the original dataset. Raw data types can differ from the cleaned types.

Column	Meaning	Data type / role	Unit / examples
Unnamed: 0	Row identifier included in the source file.	Integer; identifier	Row index / ID
Name	Car make and model as listed.	Categorical (text)	Brand and model
Location	City in which the used car is listed.	Categorical (text)	City name
Year	Year in which the car was manufactured.	Integer; numerical	Calendar year
Kilometers_Driven	Distance the car has travelled.	Numerical	Kilometres
Fuel_Type	Fuel category of the vehicle.	Categorical (text)	Diesel, Petrol, Electric
Transmission	Transmission system of the vehicle.	Categorical (text)	Manual, Automatic
Owner_Type	Ownership category in the listing.	Categorical (text)	First, Second, Third, Fourth & Above
Mileage	Reported fuel-efficiency figure.	Unit-bearing text; converted to numeric	Raw suffix should be verified before assigning one universal unit
Engine	Engine displacement/capacity.	Unit-bearing text; converted to numeric	CC
Power	Reported engine power.	Unit-bearing text; converted to numeric	bhp
Seats	Number of seats in the vehicle.	Numerical; integer-valued	Number of seats
New_Price	Reported new/showroom price.	Text/object in raw data	Price field; many missing values
Price	Listed resale price of the used car.	Numerical; target variable	Notebook plots label price in lakh; exact source unit is not confirmed in the data-understanding notes

3. Variables used in the project

Dependent variable (target): Price. This is the value the project aims to understand and eventually predict.

Potential independent variables (features): Name, Location, Year, Kilometers_Driven, Fuel_Type, Transmission, Owner_Type, Mileage, Engine, Power and Seats.

Identifier: Unnamed: 0 is a row ID, not a meaningful vehicle characteristic. The notebook removes it before analysis.

4. Changes made during data preparation

Original field	Notebook action	Reason / note
Unnamed: 0	Dropped	Identifies the row; not useful as a car feature.
Year	Renamed to Manufacturing_Year	Makes the year field's meaning explicit.
New_Price	Renamed to Showroom_Price, then dropped	The notebook reports 5,032 missing values out of 5,847.
Name, Location, Fuel_Type, Transmission, Owner_Type	Converted to string type	Makes categorical/text handling explicit.
Mileage, Engine, Power	Numeric portion extracted and converted to float	Removes unit text for numerical analysis. Engine is in CC and Power in bhp; verify Mileage suffixes in the raw data.
Seats	Converted to nullable integer type	Seat count is integer-valued; nullable integer supports missing values.

5. Data-quality observations recorded in the notebook

• The notebook reports no duplicate rows.

• Showroom_Price contains 5,032 missing values and is dropped.

• 40 zero values are identified in Mileage and replaced with missing values before imputation.

• Missing Mileage values are filled using the median for the same Name, then the median for the same Fuel_Type. Electric vehicles may retain missing Mileage because kmpl is not applicable.

• For non-electric vehicles, missing Engine values are filled using medians grouped by Name and then Fuel_Type. Missing Power values are handled similarly.

• Missing Seats values are filled using the mode for the same Name, followed by the overall mode.

• One extreme Kilometers_Driven value of 6,500,000 is treated as a likely data-entry error and replaced with the median for the same car Name.

• The notebook flags 679 Price observations as statistical outliers using the 1.5 × IQR rule. They are identified, not automatically removed.

Interpretation note: This PDF follows the supplied notebooks. The notebook's plots label Price in lakh, but the data-understanding notes do not establish the exact currency/unit from source metadata. Also verify Mileage's raw suffixes before documenting a single universal unit. A data dictionary should record uncertainty rather than disguise it.